



```
In [1]: import nltk
import pandas as pd
```

```
In [2]: df=pd.read_csv("spam.csv",encoding="latin-1")
df.head()
```

```
Out[2]:
```

	v1	v2	Unnamed: 2	Unnamed: 3	Unnamed: 4
0	ham	Go until jurong point, crazy.. Available only ...	NaN	NaN	NaN
1	ham	Ok lar... Joking wif u oni...	NaN	NaN	NaN
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	NaN	NaN	NaN
3	ham	U dun say so early hor... U c already then say...	NaN	NaN	NaN
4	ham	Nah I don't think he goes to usf, he lives aro...	NaN	NaN	NaN

```
In [3]: df.drop(["Unnamed: 2","Unnamed: 3","Unnamed: 4"],axis=1,inplace=True)
```

```
In [4]: df.head()
```

```
Out[4]:
```

	v1	v2
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

```
In [5]: df.columns=["label","text"]
```

```
In [6]: df.head()
```

```
Out[6]:
```

	label	text
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

```
In [8]: nltk.download("stopwords")
```

```
[nltk_data] Downloading package stopwords to  
[nltk_data] C:\Users\vipul\AppData\Roaming\nltk_data...  
[nltk_data] Unzipping corpora\stopwords.zip.
```

```
Out[8]: True
```

```
In [9]: import nltk  
import string  
import re  
stopwords=nltk.corpus.stopwords.words("english")
```

```
In [10]: def clean_text(text):  
    text="".join([char.lower() for char in text if char not in string.punctuat  
    tokens=re.split("\W+",text)  
    text=[word for word in tokens if word not in stopwords]  
    return text
```

```
In [13]: x="Nah I don't think he goes to usf, he lives aro?"  
x="".join([char.lower() for char in x if char not in string.punctuation])  
x
```

```
Out[13]: 'nah i dont think he goes to usf he lives aro'
```

```
In [14]: tokens=re.split("\W+",x)
```

```
In [15]: tokens
```

```
Out[15]: ['nah', 'i', 'dont', 'think', 'he', 'goes', 'to', 'usf', 'he', 'lives', 'ar  
o']
```

```
In [16]: clean_text("Nah I don't think he goes to usf, he lives aro?")
```

```
Out[16]: ['nah', 'dont', 'think', 'goes', 'usf', 'lives', 'aro']
```

```
In [17]: from sklearn.feature_extraction.text import CountVectorizer  
count_vect = CountVectorizer(analyzer=clean_text)  
X_count=count_vect.fit_transform(df["text"])  
print(X_count.shape)  
print(count_vect.get_feature_names_out())
```

```
(5572, 9395)  
[' ' '0' '008704050406' ... 'üiharry' 'ûò' 'úówell']
```

```
In [18]: X_features=pd.DataFrame(X_count.toarray())  
X_features.head()
```

```
Out[18]:
```

	0	1	2	3	4	5	6	7	8	9	...	9385	9386	9387	9388	9389	9390	9391
0	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0

5 rows × 9395 columns

```
In [19]: from sklearn.ensemble import RandomForestClassifier
In [20]: from sklearn.model_selection import train_test_split
In [21]: X_train,X_test,y_train,y_test=train_test_split(X_features,df["label"],test_size=0.2)
In [22]: rf=RandomForestClassifier()
          rf_model=rf.fit(X_train,y_train)
In [23]: y_pred=rf_model.predict(X_test)
In [24]: from sklearn.metrics import classification_report
In [25]: print(classification_report(y_test,y_pred))
```

	precision	recall	f1-score	support
ham	0.97	1.00	0.98	971
spam	1.00	0.78	0.88	144
accuracy			0.97	1115
macro avg	0.98	0.89	0.93	1115
weighted avg	0.97	0.97	0.97	1115

```
In [29]: text=["I Kill You"]
          count_vect1=count_vect.transform(text)
          X_features=pd.DataFrame(count_vect1.toarray())
          X_features.head()
Out[29]:
```

	0	1	2	3	4	5	6	7	8	9	...	9385	9386	9387	9388	9389	9390	9391
0	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0

1 rows × 9395 columns

```
In [30]: y_pred=rf_model.predict(count_vect1)
```

```
In [31]: y_pred
```

```
Out[31]: array(['ham'], dtype=object)
```

```
In [ ]:
```